

Hybrid-Auto-Scaling-Prediction-and-Monitoring-Framework

Used AWS EC2 to host backend services and run prediction scripts for auto-scaling.

Stored the frontend and datasets/model logs using AWS S3.

Used MongoDB Atlas (cloud database) for storing CloudWatch metrics and model outputs.

Integrated AWS CloudWatch to collect CPU, Memory, Disk I/O, and Network metrics every 1 minute.

Implemented Auto Scaling Groups (ASG) on EC2 to automatically scale based on demand.

Applied the AWS 30–70% CPU threshold logic for reactive stability.

Built a predictive scaling system using ML models (LightGBM, CatBoost, XGBoost).

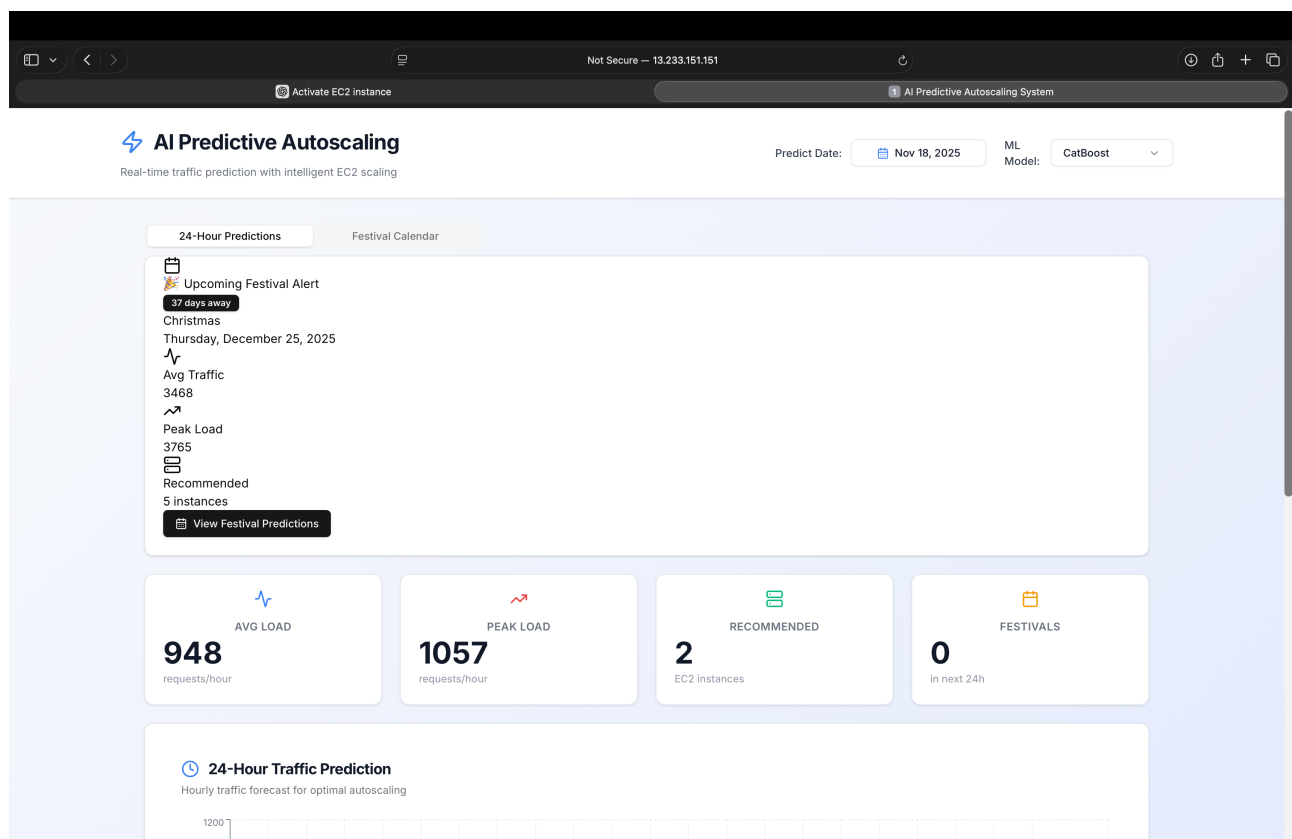
Integrated IAM Roles for secure communication between EC2, S3, CloudWatch and ASG.

Leveraged CloudFront for faster delivery of frontend files hosted on S3.

Created CloudWatch alarms for CPU/GPU thresholds to trigger scaling actions.

Added a Generative AI layer to explain scaling decisions in natural language.

Managed real-time data ingestion and prediction logs through cloud



EC2 > Instances

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Instances (5)

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public I
☐	cc-front	i-02b9aa92b70c2f46a	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-15-207-98-239.ap...	15.207.1
☐	cc-backend	i-0260b52a7242d72af	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-35-154-151-103.ap...	35.154.
☐	CCfinal-Host	i-04459fc66cc45c81b	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-13-233-151-151.ap...	13.233.
☐	CCfinal-Host	i-09a220010d7d18061	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-35-154-149-254.ap...	35.154.
☐	CCfinal-Host	i-06fec8c312b1191c1	Running	t2.micro	initializing	View alarms +	ap-south-1b	ec2-13-232-209-186.ap...	13.232.

Select an instance

CloudShell

Feedback

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Not Secure — 13.233.151.151

Activate EC2 instance

AI Predictive Autoscaling System

Hourly traffic forecast for optimal autoscaling

Traffic Load

AWS Auto Scaling Control

Trigger EC2 instance scaling based on predictions

Trigger AWS Scaling

Explain Recommendation

Real AWS

Predicted Load: 1057

Desired Capacity: 2 instances

Scaled CCfinal-ASG to 2 instances

Configuration

Selected Date: Nov 18, 2025

Model: CATBOOST

Prediction Window: 24 hours

Festival API: Calendarific

AWS Service: EC2 Auto Scaling